# MRX AND ISE Spread Top of Market Glimpse Feed

Version 2.1

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# 1. Overview

A complement to the MRX AND ISE Spread Top of Market real-time data feed product, MRX AND ISE GLIMPSE for Spread Top of Market is a point-to- point data feed connection that provides direct data feed customers with the current Top of Book in the MRX AND ISE execution system. MRX AND ISE GLIMPSE for Spread Top of Market uses the same data formats as the MRX AND ISE Spread Top of Market data feed product.

# 2. Architecture

MRX AND ISE GLIMPSE for Spread Top of Market (MRX AND ISE Top)is a point-to-point data feed product comprised of a series of sequenced messages. Each message is variable in length based on the message type. The messages that make up the MRX AND ISE GLIMPSE for Spread Top of Market (MRX AND ISE Spread Top) protocol are typically delivered using a higher level protocol that takes care of sequencing and delivery guarantees.

MRX AND ISE currently offers the Options GLIMPSE data feed in the SoupBinTCP protocol option only. Please note that GLIMPSE users must login to SoupBinTCP for sequence 1 to correctly receive data.

# 3. Data Types

All Alpha or Alphanumeric fields are left justified and padded on the right with spaces.

All Integer fields are unsigned big-endian (network byte order) binary encoded numbers unless otherwise specified. Integers may be 1, 2, 4 or 8 bytes long.

Prices are 2 byte or 4 byte Integer fields. 2 byte Price fields are unsigned positive numbers. 4 byte Price fields are signed numbers. When a 4 byte price is converted to a decimal format, prices are in fixed point format with 6 whole number places followed by 4 decimal digits. When a 2 byte price is converted to a decimal format, prices are in fixed point format with 3 whole number places followed by 2 decimal digits.

Negative prices will be indicated with a negative integer at the start of the price field.

# 4. Message Formats

This feed supports the following types of messages:

- System Events
- Complex Strategy Directory Messages
- Strategy Trading Action Messages
- Best bid and offer updates
- Glimpse Snapshot message

Within the system event and administrative types, the options system may support multiple message formats as outlined below.

## 4.1. System Event Message

The system event message type is used to signal a market or data feed handler event. The format is as follows:

**System Event Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | “S” = System Event Message |
| Tracking Number | 1 | 2 | Integer | Internal system tracking number |
| Timestamp | 3 | 8 | Integer | Nanoseconds since midnight. |
| Event Code | 11 | 1 | Alpha | Refer to System Event Codes below |

**System Event Codes**

| Code | Explanation | When (typically) |
|---|---|---|
| “O” | Start of Messages. This is always the first message sent in any trading day. | After ~12:30am |
| “S” | Start of System Hours. This message indicates that the options system is open and ready to start accepting orders. | After Start of Messages and before Start of Currency Opening Process |
| “Q” | Start of Opening Process. This message is intended to indicate that the options system has started its opening auction process. | 9:30:00am |
| “N” | Start of Normal Hours Closing Process. This message is intended to indicate that the options system will no longer generate new executions for options that trade during normal hours. | 4:00:00pm |
| “L” | Start of Late Hours Closing Process. This message is intended to indicate that the options system will no longer generate new executions for options that trade during extended hours. | 4:15:00pm |
| “E” | End of System Hours. This message indicates that the options system is now closed. | ~5:15pm |
| “C” | End of Messages. This is always the last message sent in any trading day. | ~5:20pm |
| “W” | End of WCO Early closing. This message is intended to indicate that the exchange will no longer accept any new orders or changes to existing Orders on last trading date of WCO options. | 12:00 Noon |

## 4.2. Administrative Messages

### 4.2.1. Complex Strategy Directory Message

Whenever a complex order is added in the system for an underlying, the order is normalized and results in either the creation of a new complex strategy or is added to an existing strategy. A Complex Strategy Directory Message containing the strategy definition will be sent. For GTC strategies, these will be assigned each trading day and will not be persistent across trading days. The Strategy ID assigned for a new complex strategy is unique for a particular complex instrument for a trading session. However, Strategy IDs are independent of session Options IDs and uniqueness of the IDs across both complex and simple options is not guaranteed.

- The Underlying Symbol is in most cases the same as the industry standard ticker underlying. In rare cases, such as a special settlement symbol, the exchange assigns unique underlying symbols.

**Complex Strategy Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | "s" = Directory Message |
| Tracking Number | 1 | 2 | Integer | Internal System Tracking Number |
| Timestamp | 3 | 8 | Integer | Nanoseconds since midnight |
| Strategy ID | 11 | 4 | Integer | Option ID assigned daily. Valid for trading day. |
| Strategy Type | 15 | 1 | Alphanumeric | “V” = Vertical Spread<br>“T” = Time Spread<br>“D” = Diagonal Spread<br>“S” = Straddle<br>“G” = Strangle<br>“C” = Combo<br>“R” = Risk Reversal “A” = Ratio Spread<br>“B” = Box Spread<br>“F” = Butterfly Spread<br>“U” = Custom |
| Underlying Symbol | 16 | 13 | Alphanumeric | Underlying Symbol for the strategy. All legs in this strategy belong to this Underlying |
| Reserved | 29 | 16 | Alpha | Reserved for future use. |

**Complex Strategy Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Number of legs | 45 | 1 | Integer | Number of legs in the strategy Legs NOTE: Leg field offsets below are an equation, where "n" is the zero based leg number (0, 1, ...) |
| Leg Information, legs repeated. N=0, Option ID | 25n + 46 | 4 | Integer | Option ID for this leg, valid for the trading day. The same ID as the corresponding Option in the Options Directory Message. Zero (0) for Stock Leg. |
| Security Symbol | 25n + 50 | 8 | Alphanumeric | Denotes the option root symbol (security symbol) |
| Expiration Year | 25n + 58 | 1 | Integer | Last two digits of the year of the option expiration |
| Expiration Month | 25n + 59 | 1 | Integer | Expiration Month of the option (1-12) |
| Expiration Day | 25n + 60 | 1 | Integer | Day of the Month of expiration (1-31) |
| Explicit Strike Price | 25n + 61 | 4 | Integer | Explicit strike price. Refer to Data Types for field processing notes. Zero (0) for Stock Leg. |
| Option Type | 25n + 65 | 1 | Alpha | "C" = Call option<br>"P" = Put option<br>Blank (" ") for Stock Leg. |
| Side | 25n + 66 | 1 | Alpha | Indicates the side of the leg:<br>"B" = Leg is on Buy side<br>"S" = Leg is on Sell side |
| Leg Ratio | 25n + 67 | 4 | Integer | Leg Ratio |

### 4.2.2. Strategy Trading Action Message

The options system uses this administrative message to indicate the current trading status of an index or equity option within the options market.

Whenever a strategy is created and assigned a Strategy ID, MRX AND ISE will send a Strategy Trading Action Message with current trading state "T" (Trading) or "H" (Halted) soon after the Complex Strategy directory message is sent. Trading firms should assume that all strategies are eligible for trading. MRX AND ISE will send out a Trading Action message with "H" (HALTED) when a strategy is halted for trading. Thereafter throughout the trading day the Trading Action message is used to relay changes in trading status for the strategy. Messages will be sent when the strategy is halted or is released for trading.

**Trading Action Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | "H" = Trading Action |
| Tracking Number | 1 | 2 | Integer | Internal system tracking number |
| Timestamp | 3 | 8 | Integer | Nanoseconds since midnight |
| Strategy ID | 11 | 4 | Integer | Integer ID of the Strategy, as defined in the Complex Strategy Message. |
| Current Trading State | 15 | 1 | Alphabetic | Reflects the current trading state for the State derivative. The allowable values are:<br>"H" = Halt in effect<br>"T" = Continuous Trading<br>"I" = Pre-Open<br>"O" = Opening Auction<br>"R" = Re-Opening<br>"X" = Closed |

## 4.3. Strategy Best Bid AND Ask Update

The options system will continuously calculate its best bid and ask position for active strategies on the options market during the trading day. Whenever the best bid and ask position changes on both sides, the options system will send its best bid and ask update via the data feed for the affected security. A change in bid or ask implies a change in price and/or sizes. The quote will reflect the highest price displayable in the options system for buy orders and the lowest price displayable in the options system for sell orders. Customer and ProCust sizes are displayed as well.

If only one side of the quote changes, Best Bid OR Ask Update message will be sent for bandwidth efficiency reasons. The Quote Condition applies to both the bid and ask sides.

**Best Bid AND Ask Update**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | “E” = Best bid AND ask update |
| Tracking Number | 1 | 2 | Integer | Internal system tracking number |
| Timestamp | 3 | 8 | Integer | Nanoseconds since midnight |
| Strategy ID | 11 | 4 | Integer | Integer ID of the strategy, as defined in the Strategy Directory Message. |
| Quote Condition | 15 | 1 | Alpha | <space> = regular quote/autox eligible |
| Bid Market Size | 16 | 4 | Integer | The aggregate market size on Bid side. |
| Bid Price | 20 | 4 | Integer | Best bid price in fixed point format with 6 whole number places followed by 4 decimal digits |
| Bid Size | 24 | 4 | Integer | Aggregated number of contracts on the bid side being displayed in the options market at the current time. Does not include the market size. |
| Bid Cust Size | 28 | 4 | Integer | Customer quantity on the bid side. |
| Bid ProCust Size | 32 | 4 | Integer | Customer professional quantity on the bid side. |
| Bid DNTT Size | 36 | 4 | Integer | Aggregated Bid DNTT (Do not Trade Through) Limit Size |
| Bid DNTT Market Size | 40 | 4 | Integer | The aggregated DNTT (Do Not Trade Through) Market Size on Bid Size. |
| Ask Market Size | 44 | 4 | Integer | Aggregate market quantity on Ask side. |
| Ask Price | 48 | 4 | Integer | Best ask price in fixed point format with 6 whole number places followed by 4 decimal digits |
| Ask Size | 52 | 4 | Integer | Aggregated number of contracts on the ask side being displayed in the options market at the current time. Does not include the market size. |
| Ask Cust Size | 56 | 4 | Integer | Customer quantity on the ask side. |
| Ask ProCust Size | 60 | 4 | Integer | Customer professional quantity on the ask side. |
| Ask DNTT Size | 64 | 4 | Integer | Aggregated Ask DNTT (Do not Trade Through) Limit Size |
| Ask DNTT Market Size | 68 | 4 | Integer | The aggregated DNTT (Do Not Trade Through) Market Size on Ask Size. |

## 4.4. Strategy Best Bid OR Ask Update

The options system will continuously calculate its best bid and offer position for active strategies on the options market during the trading day. Whenever the best bid or ask position changes on one side but not the other side, the options system will send its best bid or ask update via this feed for the affected security. A change in bid or ask implies a change in price and/or size. The quote will reflect the highest price displayable in the options system for buy orders/quotes and the lowest price displayable in the options system for sell orders/quotes.

For the bid or ask update received, the data feed recipient firm should adjust the quotation bid or ask side only for the market side indicated in the message. Implicitly the opposite side has the same price and size as previously displayed.

If both bid and ask change as one update, the Best Bid AND Ask Update message will be sent, displaying both sides of the quote simultaneously.

**Best Bid OR Ask Update Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | Strategy Best bid OR ask update:<br>“c” = Quote update bid side<br>“d” = Quote update ask side |
| Tracking Number | 1 | 2 | Integer | Internal system tracking number |
| Timestamp | 3 | 8 | Integer | Nanoseconds since midnight |
| Strategy ID | 11 | 4 | Integer | Integer ID of the strategy, as defined in the Strategy Directory Message. |
| Quote Condition | 15 | 1 | Alpha | <space> = regular quote/autox eligible |
| Market Size | 16 | 4 | Integer | The aggregate market size on Bid or Ask side. |
| Price | 20 | 4 | Integer | Best bid or ask price in fixed point format with 6 whole number places followed by 4 decimal digits |
| Size | 24 | 4 | Integer | Aggregated number of contracts on the bid or ask side being displayed in the options market at the current time. Does not include the market size. |
| Cust Size | 28 | 4 | Integer | Customer quantity on the bid or ask side. |
| ProCust Size | 32 | 4 | Integer | Customer professional quantity on the bid or ask side. |
| DNTT Size | 36 | 4 | Integer | Aggregated DNTT (Do Not Trade Through) Limit Size |
| DNTT Market Size | 40 | 4 | Integer | The aggregated DNTT (Do Not Trade Through) Market Size on Bid or Ask Size. |

## 5. Glimpse for Top of Market Spread Feed

The Snapshot message reflects the MRX AND ISE Options Spread Top of Market (MRX AND ISE Top) sequence number at the time that the MRX AND ISE Options GLIMPSE for MRX AND ISE Spread Top of Market spin was requested.

To maintain a real-time order display, firms should begin to process real-time MRX AND ISE Options Top of Market messages beginning with the sequence number stated in this MRX AND ISE Options GLIMPSE for MRX AND ISE Top of Market snapshot message

**Snapshot Message**

| Name | Offset | Length | Value | Notes |
|---|---|---|---|---|
| Message Type | 0 | 1 | Alpha | “M” = End of Snapshot Message |
| Sequence Number | 1 | 20 | Numeric | Top of Market Spread Feed sequence number when the Glimpse snapshot was taken. To keep the stream current, process the Top of Market Spread Feed messages beginning with the message sequence number in this snapshot message.<br>Note: While Top of Market Spread Feed is a binary data feed, the<br>SoupBINTCP protocol uses ASCII characters for the sequence number in the logon request message format. |

## 6. Support

- For general product support for Nasdaq data feeds, please contact Nasdaq Market Data at Clientsuccess@nasdaq.com.

## 7. Document History

| Revision | Published | Author(s) | Summary of Changes |
|---|---|---|---|
| Version 2.0 | March 16, 2022 | | • Initial Draft |
| Version 2.01 | June 16, 2022 | | • Complex Strategy Message Updates:<br>– Butterfly Spread enumeration from “Y” to “F” in the Strategy Field<br>– Notes to Expiration Month Field<br>– Notes to Explicit Strike Price Field to indicate Zero (0) for Stock Leg<br>– Removed “N” = N/A enumeration from Option Type Field and added Blank (“ “) = Stock Leg<br>– Correction to offsets starting with Explicit Strike Price Side fields<br>• Correction to Bid NTT Market Size and Ask NTT Market Size Field name to DNTT for Do Not Trade Through on Best Bid AND Ask Update Message and Best Bid OR Ask Update Message. Notes updated accordingly. |
| Version 2.02 | January 9, 2023 | | • Version updated to 2.02<br>• Clarifying the Data Types: Added “2 byte Price Fields are unsigned positive numbers. 4 byte Price fields are signed positive numbers.” |
| Version 2.02 | April 6, 2023 | | • Corrected the Data Types: Changed “4 byte price fields are signed positive numbers” to “4 byte price fields are signed numbers” |
| Version 2.02 | April 1, 2024 | | • ISE will adopt Spread Top of Market Glimpse v2.02 during the replatform scheduled to complete September 23,2024 |
| Version 2.1 | October 2, 2024 | | Complex Strategy Directory Message<br>• Message type changed to "s"<br>• Increase symbol size from 6 characters to 8 characters<br>• Add reserved field of 16 characters |
| Version 2.1 | January 9, 2025 | | Appendix A<br>• Flex Symbology table added |

# Appendix A

## Flex Symbology

| Underlying Security | Lead Character | Settlement Type | Exercise Style |
|---|---|---|---|
| Index | 1 | AM | American |
| Index | 2 | AM | European |
| Index | 3 | PM | American |
| Index | 4 | PM | European |
| Equity/ETF | 1 | PM | American |
| Equity/ETF | 2 | PM | European |
| ETF Cash | 3 | PM | American |
| ETF Cash | 4 | PM | European |

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